Problem 1

1. On the Gisette dataset, train decision trees of maximum depth 1,2,$\cdots$ up to 12, for a total of 12 decision trees. Use the trained trees to predict the class labels on the training and test sets and obtain the training and test misclassification errors. Plot on the same graph the training and test misclassification errors vs tree depth as two seperate curves. Report in a table the minimum test error and the tree depth for which the minimum was attained.

   We do this in Python, and obtain a plot of the training and test misclassification errors vs tree depth in figure 1:

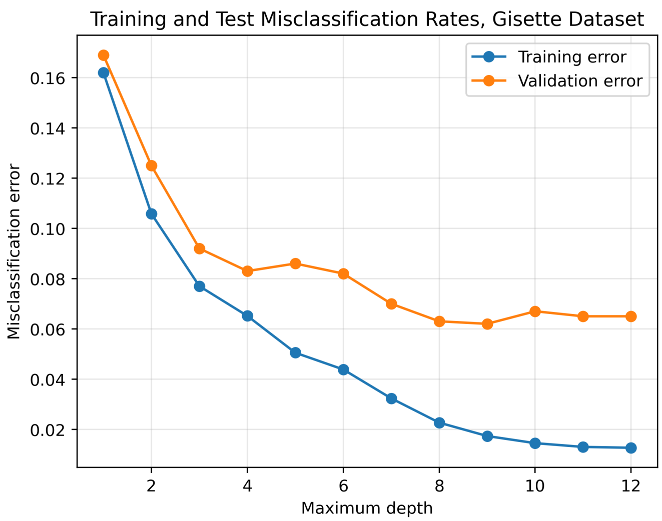

Figure 1: Misclassification Error vs Tree Depth, Training and Validation: Gisette dataset

   The table freporting the minimum test error and the tree depth for which the minium was attained are displayed in table 1.

Depth	Training Misclassification	Validation Misclassification
9	0.017500	0.061000

Table 1: Lowest misclassification error, Gisette Dataset

2. Repeat point 1.) on the satimage, madelon, hill-valley, and covtype datasets.

   We can do this also, and obtain the four images shown in figure 2.

   The table reporting the minimum test error and tree depth for the various datasets is shown in table 2

3. On the Gisette dataset, for each off $k \in \{3, 10, 30, 100, 300\}$ train a random forest with k trees where the split attribute at each node is chosen from a random subset of $71 \approx \sqrt{5000}$ features. Use the trained trees to predict the class labels on the training

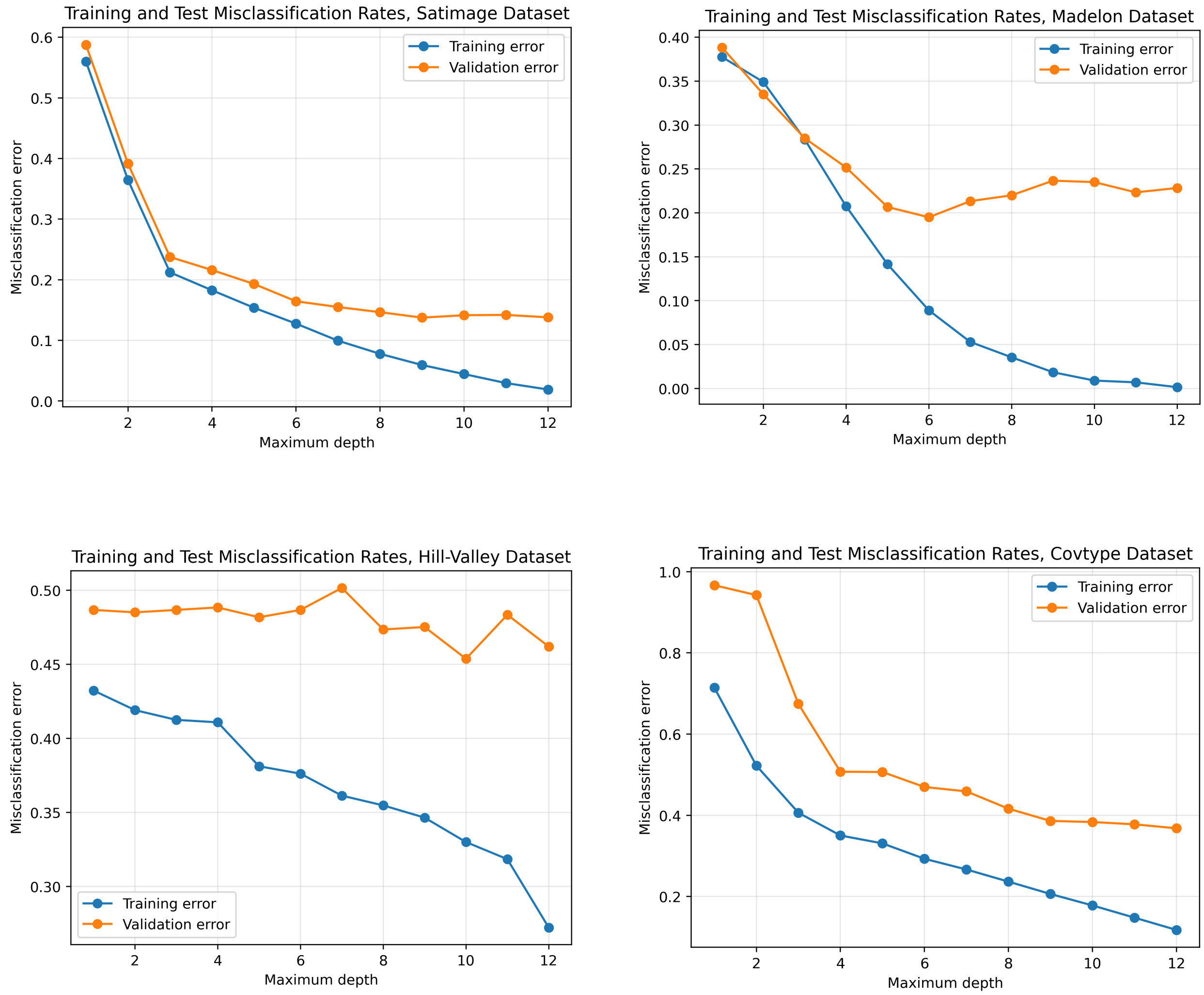

Figure 2: Training and Validation datasets misclassification error rates for four datasets

and test sets and obtain the training and test misclassification errors. Plot on the same graph the training and test errors vs. number of trees k as two seperate curves. Report the training and test misclassification errors in a table.

We do this in Python, and obtain the plot of the training and test misclassification errors vs tree depth shown in figure 3:

The table reporting the training and test errors for various random forests are shown in table 3:

4. Repeat point 3.) on the satimage, madelon, and hill-valley datasets.

   The images showing the training and test misclassification errors are displayed in figure 4.

5. Repeat point 3.) on the Gisette dataset where the split attribute at each node is chosen from a random subset of $12 \approx \log_2(5000)$ features, and where the split attribute at

Dataset	Depth	Training Misclassification	Validation Misclassification
Satimage	9	0.059076	0.138000
Madelon	6	0.089000	0.201667
Hill-Valley	8	0.354785	0.465347
Covtype	12	0.117659	0.368692

Table 2: Lowest misclassification error across datasets

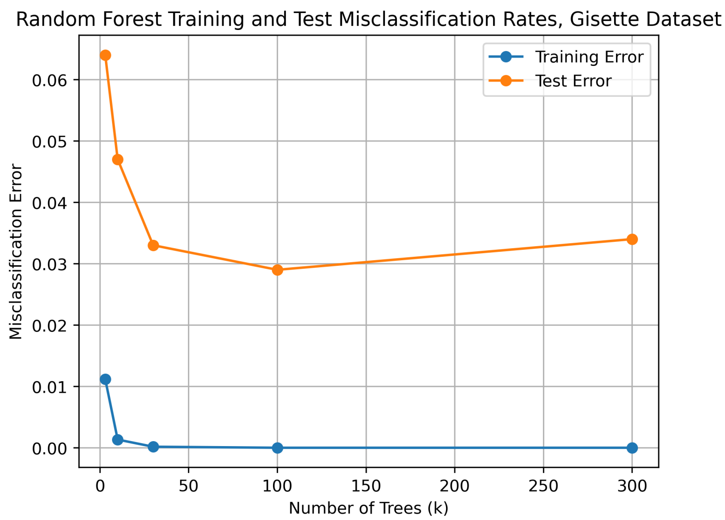

Figure 3: Misclassification Error vs Tree Depth, Training and Validation: Gisette dataset

each node is chosen from all 5000 features, plotting all four curves (2 train and 2 test misclassification errors) on the same graph.

Number of Trees	Training Misclassification Error	Validation Misclassification Error
3	0.011167	0.064000
10	0.001333	0.047000
30	0.000167	0.033000
100	0.000000	0.029000
300	0.000000	0.034000

Table 3: Misclassification errors, Random Forests, Gisette Dataset

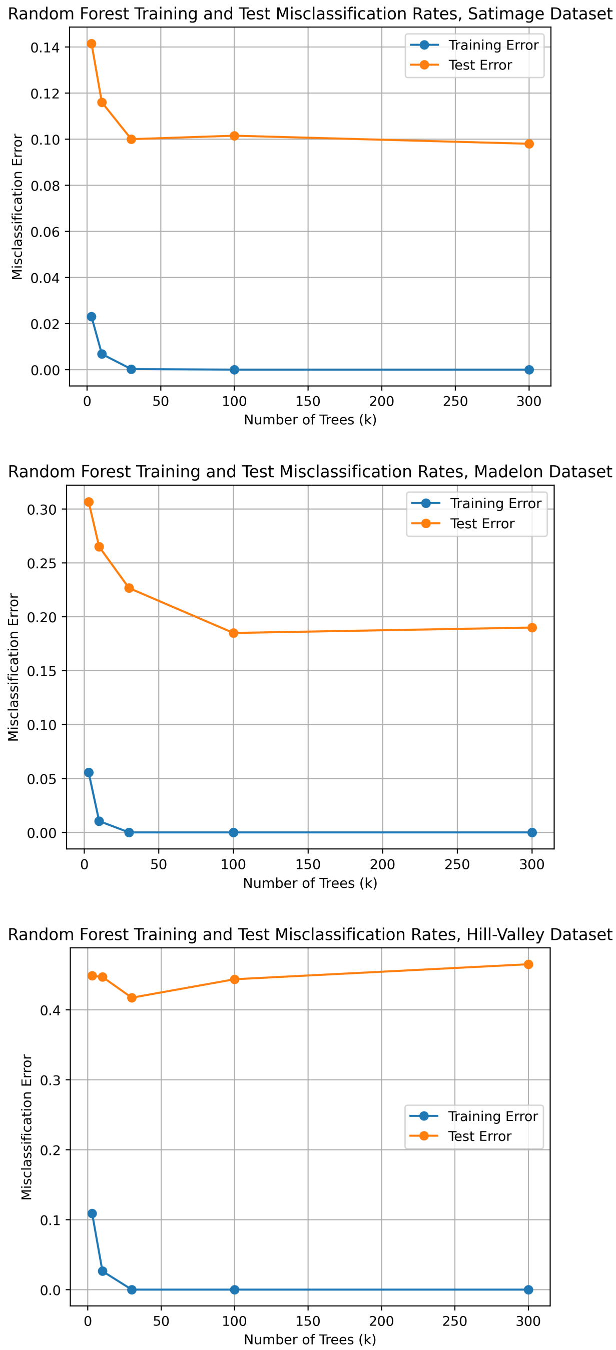

Figure 4: Plots Showing training and test misclassification errors vs number of trees.